RCM steps:	Step 1		Step 2	Step 3			Step 4			Step 5				Step 6			Step 7									
System	Equipment	Function	Functional Failure	Failure mode	Failing component	Asset number	Asset description	Failure cause	Failure Effect	Evident (E) vs. Hidden (H) Safety/environment (S) vs. Financial (F)	CoF	PoF (MTBF)	Risk	Proactive tasks	Frequency	Corrective Task	Comment	Design min	design max	normal	Design unit	Operating min	Operating max	Normal	Operating unit	Timseries
23 Gas compression	**23-HA-9115** 1st stage discharge cooler Heat Exchanger Shell _ tube	**Main function:** To exchange heat between the warm gas from the gas compressor and the cooling medium system to keep the gas within 10% of its operating temperature. **Subfunction:** To contain process and utility mediums so they do not mix or leak. **Operating conditions:** Shell temp: 20.12 deg C Shell pressure: 5 barg Tube temp: 119.7 deg C Tube pressure: 12 barg	**MF Failure:** Unable to cool down gas to within 10% of its operating temperature (108 < temp < 132) **SF Failure:** Unable to contain the mediums	AIR - Abnormal instrument reading	Instrument / transmitter	23-TT-92602	1STSTGDISCCOOL GAS IN	Instrument / transmitter failure or lack of calibration.	Failure to properly control heat thrasifer due to incorrect flow rates of process or utility medium. Potential for HH or LL	EF1	4 - High	c - Medium	H	Monitoring instrument readings and comparing agains thresholds, both HEX - mounted and system mounted		Calibration										
						23-TT-92603	1STSTGDISCCOOL GAS OUT											70	84	77	Celcius	56	72			VAL_23-TT-92603:X.Value
						23-PDT-92602	1ST STG DISCH GAS COOLERS															0.3	0.9		barg	VAL_23-TT-92602:X.Value
						23-FT-92537-01	1STSTGCOMP DISCHARGE															164	294		(DQ) 0-27m3/hr)	VAL_23-FT-92537-01:X.Value
						23-FT-92537-02	1STSTGCOMP DISCHARGE															100	250			VAL_23_FT_92537_02.Z.X.Value
						45-TT-92607A	1STSTGDISCCLR COOLMED OUT											157.5	192.5	175	Celcius	97.2	118.2	108	Celcius	VAL_45-TT-92607A:X.Value
						45-TT-92607B	1STSTGDISCCLR COOLMED OUT																		Celcius	VAL_45-TT-92607B:X.Value
						45-FT-92606	1STSTGDISCCOOL COOLMED IN															70	110	100	124Am3/h +/-10%	VAL_45-FT-92606:X.Value
						45-TT-92606	1STSTGDISCCLR COOLMED SPLY															15	19	16.5	Celcius	VAL_45-TT-92606:X.Value
						45-PT-92608	1STSTGDISCCOOL CM OUT															4	6	5		VAL_45-PT-92608:X.Value
						45-PT-92609	1STSTGDISCCOOL CM OUT											4	16	14	barg	5	14	barg		VAL_45-PT-92609:X.Value
						45-PDT-92606	1ST STG DISCH GAS COOLERS															0.4	0.6	0.5		VAL_45-PDT-92606:X.Value
				ELP - External leakage - process medium	Process pressure boundar	23-HA-9115	VRD - 1ST STAGE DISCHARGE COOLER 2	Corrosion, cracking, fatigue, damaged/corroded gaskets or mechanical looseness of joints in process piping	Release of gas or oil, leading to gas detection and shutdown, potential for fire.	HS4	4 - High	b - Low	M	External inspection	2 Y		Robot operator rounds						0	5		barg	
				ELU - External leakage - utility medium	Utility pressure boundary	23-HA-9115	VRD - 1ST STAGE DISCHARGE COOLER 2	Corrosion, cracking, fatigue, damaged/corroded gaskets or mechanical looseness of joints in utility piping	Small leak - little consequence. Large leak - loss of ability to heat/cool and potential for environemnt damage based on utility medium.	HS4	4 - High	b - Low	M	External inspection	2 Y		Robot operator rounds						0	5		barg	
				IHT - Insufficient heat transfer	Not Specified	23-HA-9115	VRD - 1ST STAGE DISCHARGE COOLER 2	Too low flow of utility medium. Too high gas temperature. Too high gas flow. Wrong temperature of utility medium. Inadequate design for experienced operating conditions	Out of tolerance temperatures on discharge side	EF1	5 - Very High	d - High	VH	Monitor heat exchanger effect in relation to input medium conditions, flow, etc.		Cleaning (off-line) NDT - Eddy Current						0.000000399	0.000000449		m2K/W	23-IHT-00345_ONS
				INL - Internal leakage	Tubes & tube-plate	23-HA-9115		Damage to tubes & tube-plate.	Process medium transfer to utility medium Utility medium transfer to process medium Potential overpressure on utility side	EF1	4 - High	c - Medium	H	Internal inspection Pressure test	5 Y 10 Y	NDT - Phased array Tube replacement								12 barg	
				PDE - Parameter deviation	Flow control valve	45-TT-92606	1STSTGDISCCOOL COOLMED IN	Input temperatures out of range for process and/or utility medium. Flow obstruction preventing heat transfer. Control failure giving inadequate / excessive flow in process or utility medium.	Excessively hot/cold process medium may damage HEX materials and not be adequately cooled / heated due to insufficient heat transfer from out-of-spec inlet temperature	EF1	3 - Medium	c - Medium	M	FCV position in relation to temperatures												
				PLU - Plugged / Choked	Tubes	23-HA-9115		Plugging of tubes or filling of shell side due to algae / debris deposits in the system.	Reduced flow of process or utility medium leading to inadequate heat transfer.	EF1	4 - High	c - Medium	H	Monitor Delta-P and flow rate.		Chemical cleaning (on-line) Pigging Tube replacement										
				SER - Minor in-service problems	Not Specified	23-HA-9115				EF1	1 - Very Low	b - Low	L	Outlet temperature deviation in relation to HM/CM temperature can indicate fault.			Robot operator rounds								NA	
				STD - Structural deficiency	Supports. Pressure boundary	23-HA-9115		Corrosion, cracking. Fluid erosion. Mechanical damage	Leak of process or utility fluids/gases through cracks or flanges.	HS4	5 - Very High	a - Very Low	M	External inspection	2 Y		Robot operator rounds								NA	
				UNK - Unknown	Not Specified	23-HA-9115				EF1	1 - Very Low	a - Very Low	VL				Robot operator rounds								NA	